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BCA(Sem-VI)-Data

Mining (603) DSE-03

2024

Time : 3 hours

Full Marks : 70

*Candidates are required to give their answers
in their own words as far as practicable.*

The figures in the margin indicate full marks.

*Answer from **all** the section as directed.*

Section-A

Multiple choices questions

(Compulsory)

1. Choose the correct answer from the following
multiple choice questions : $2 \times 10 = 20$

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(1)

(Turn over)

- (i) Data mining is the process of :
- (a) Data collection
 - (b) Data cleansing
 - (c) Extracting patterns from large data sets
 - (d) Data storage
- (ii) The process of discovering hidden patterns in large data sets is known as :
- (a) Data warehousing
 - (b) Data mining
 - (c) Data modeling
 - (d) Data integration
- (iii) Which one of the following is not a component of a data warehouse architecture ?
- (a) Data source
 - (b) Data warehouse

- (c) OLAP servers
- (d) DBMS
- (iv) Which of the following is a key feature of a data warehouse ?
 - (a) Real time data processing
 - (b) Transactional data storage
 - (c) Subject oriented data
 - (d) Single level data storage
- (v) Which of the following is a data mining functionality ?
 - (a) Data preprocessing
 - (b) Data summarization
 - (c) Classification
 - (d) Data modeling
- (vi) Which data mining technique is used to predict continuous values ?
 - (a) Clustering

- (b) Association rule mining
 - (c) Regression
 - (d) Classification
- (vii) What does OLAP stands for in data warehousing ?

- (a) Online analytical processing
- (b) Online algorithmic processing
- (c) Online application processing
- (d) Online Asynchrnous processing

(viii) The process of grouping a set of objects into classes of similar objects is known as :

- (a) Classification
- (b) Clustering
- (c) Regression
- (d) Summarization

(ix) In data warehousing, ETL stands for :

(a) Extract, Transform, Load

(b) Evaluate, Test, Log

(c) Execute, Transfer, Link

(d) Export, Translate, List

(x) Which of the following is an issue related to data quality in data mining ?

(a) Data redundancy

(b) Data inconsistency

(c) Data format

(d) Data storage capacity

Section-B

Short answer type questions

Answer any **four** questions of the following :

5×4=20

2. What is data warehouse ? Differentiate between data mining and data warehouse.

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(5)

(Turn over)

3. Define data integration. Explain three approaches of data integration.
4. Define data transformation. Explain three data transformation techniques.
5. Explain the challenge of scalability in data mining. What strategies can be employed to handle large data sets effectively ?
6. What is clustering in data mining ? Explain how clustering is different from classification.
7. Explain the concept of association rule mining.

Section-C

Long answer type questions

Answer any **two** questions of the following :

15×2=30

8. Define data mining. Explain the application of data mining.

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(6)

Contd.

9. What is data cleaning ? Explain the steps of data cleaning.
10. Explain the overall architecture of a data mining system with a suitable figure.
11. Discuss the issue of data privacy in data mining. How can data mining techniques be designed to protect sensitive.

————— x —————